ENG-0019: System Postmortem for reporting-worker

Impact

The reporting-worker system, designated with system ID SYS-049, is a mission-critical component within Project Ember. Its operational status directly influences core reporting functionalities across the organization. The recent incident, INC-009, caused a significant disruption due to a network partition, resulting in an extended resolution time of 69.5 hours. This downtime potentially affected data availability and reporting accuracy, emphasizing the importance of maintaining system reliability.

Timeline

- The incident INC-009 was reported by E-194 and involved a network partition as the root cause.
- The incident was identified and escalated promptly, with resolution efforts commencing immediately.
- The resolution was completed after 69.5 hours, restoring full system functionality.

Root Cause

The root cause of INC-009 was identified as a network partition within the infrastructure supporting reporting-worker. This network partition hindered communication between critical system components, leading to system unavailability. The issue was traced back to underlying network infrastructure failures, which compromised the system's ability to operate normally.

Action Items

- Conduct a comprehensive review of the network infrastructure supporting reporting-worker to prevent recurrence of similar network partitions.
- Implement enhanced monitoring and alerting mechanisms to detect network issues proactively.
- Schedule regular infrastructure audits, focusing on network resilience and redundancy.
- Ensure that the on-call employee, Stephanie Ross, is briefed on incident response procedures related to network failures, given her role as the on-call engineer.
- Document lessons learned and update incident response protocols accordingly.

Additional Context

The reporting-worker system is owned by Project Ember, which was completed on the start date of 2024-10-09. The project is managed by Christine Wright, a Staff Engineer based in Sydney, with her team overseeing the system's lifecycle. The on-call engineer during the incident was Stephanie Ross, an Accountant located in Melbourne, with email stephanie.ross@orbitalcorp.com.au. The system utilizes a Java/Spring/Oracle tech stack and is classified as mission-critical, underscoring the importance of robust infrastructure and rapid incident response.

This postmortem aims to improve system resilience and ensure minimal disruption to reporting services in future operations.